## 3.1 SYSTEM FLOW

The logical workflow of the Emotion-Based Music Recommendation System is designed to ensure smooth interaction between the user, the emotion recognition model, contextual data modules, and the recommendation engine. As illustrated in **Figure 3.1 (System Flow Diagram)**, the system processes the user's input and relevant external data in a sequential manner to generate accurate, real-time music recommendations that reflect the user's current emotional and environmental state.

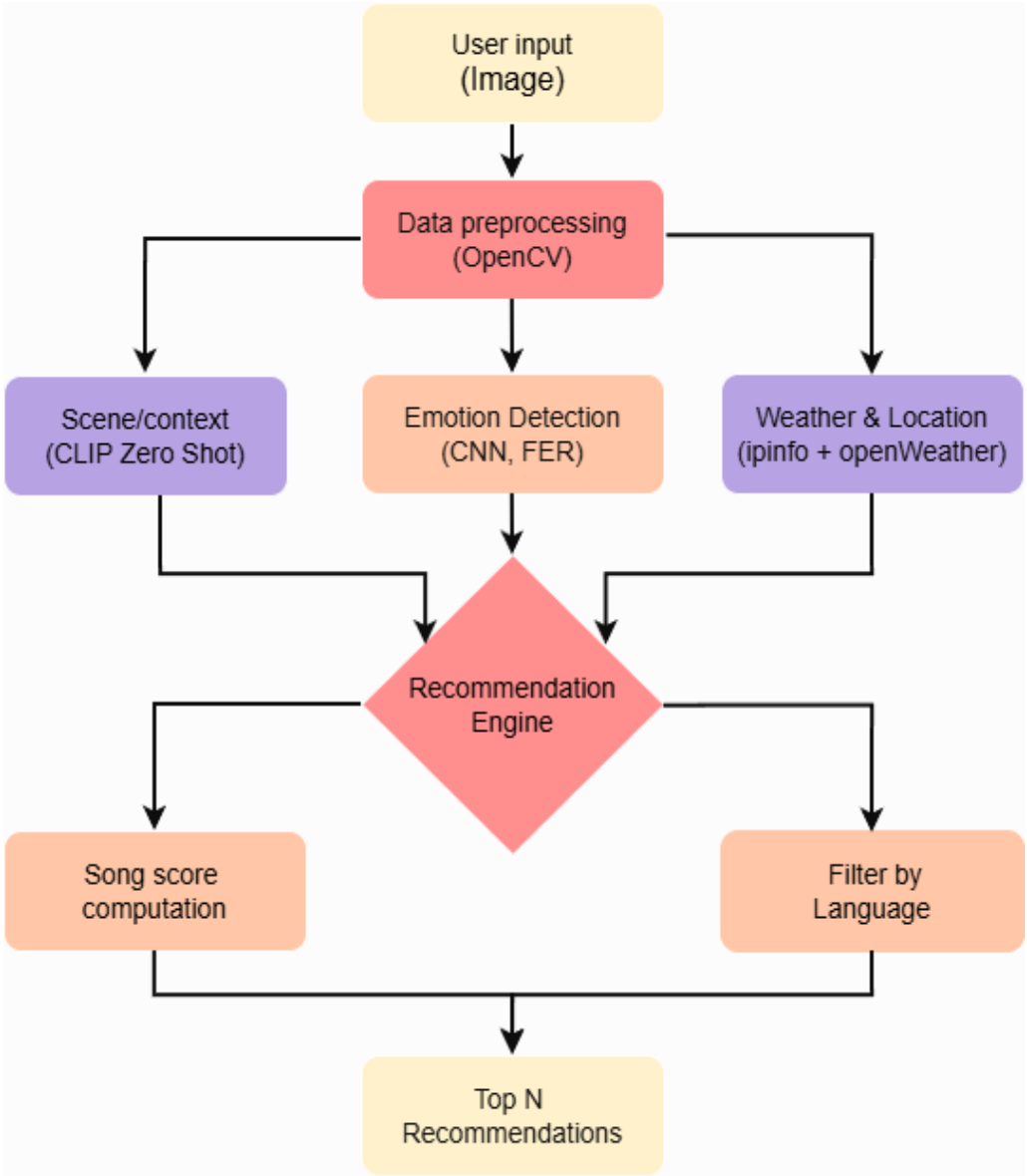


**Figure 3.1:** System Flow Diagram

**Fully Connected (Dense) Layers**

- Dense layers integrate the extracted features and learn higher-level representations of facial expressions.
- Additional **dropout** may be applied to improve regularization and reduce overfitting.

**Output Layer**

- Softmax is applied to output the likelihood of each emotion category, as described in **(Equation 3.4: Softmax classifier),** which converts raw logits into normalized probability values.

$$P(y = i|x) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}} \qquad \text{.....(3.4)}$$

- where K represents the total number of emotion categories.
- Here, $z_i$ denotes the logit value for class i, x is the input image, and $\Sigma_j$ e^($z_j$) serves as the normalization term computed over all K emotion classes. This operation transforms the raw logits into probabilities.
- The layer contains one neuron per emotion class.

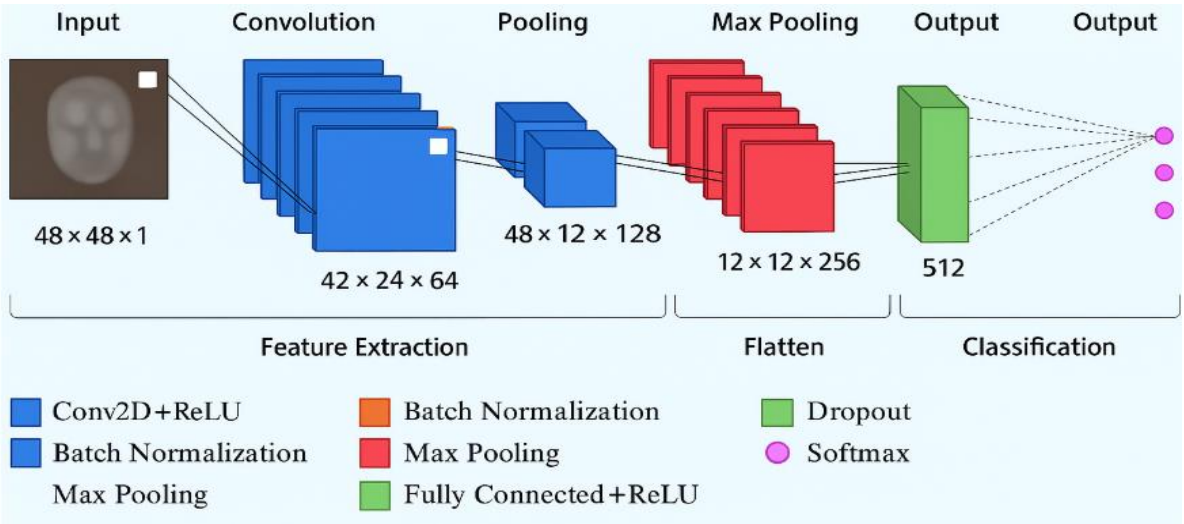


**Figure 3.2:** Facial Emotion Model Architecture

## 4.1 CONFUSION MATRIX ANALYSIS

The confusion matrix offers a clear breakdown of the model's ability to recognize each emotion by mapping its predictions against the true labels. Correct classifications appear along the diagonal, while any misplaced predictions reveal where certain emotions were mistaken for others. As illustrated in **Figure 4.1**, the matrix demonstrates that the model performs reliably across the primary emotion categories, showing that it effectively learns and distinguishes the facial cues associated with states like joy, sadness, anger, and neutral expressions. Although a small amount of misclassification is observed particularly among visually similar emotions such as fear and surprise—the overall distribution indicates that these errors remain minimal and fall within acceptable limits for real-time applications.

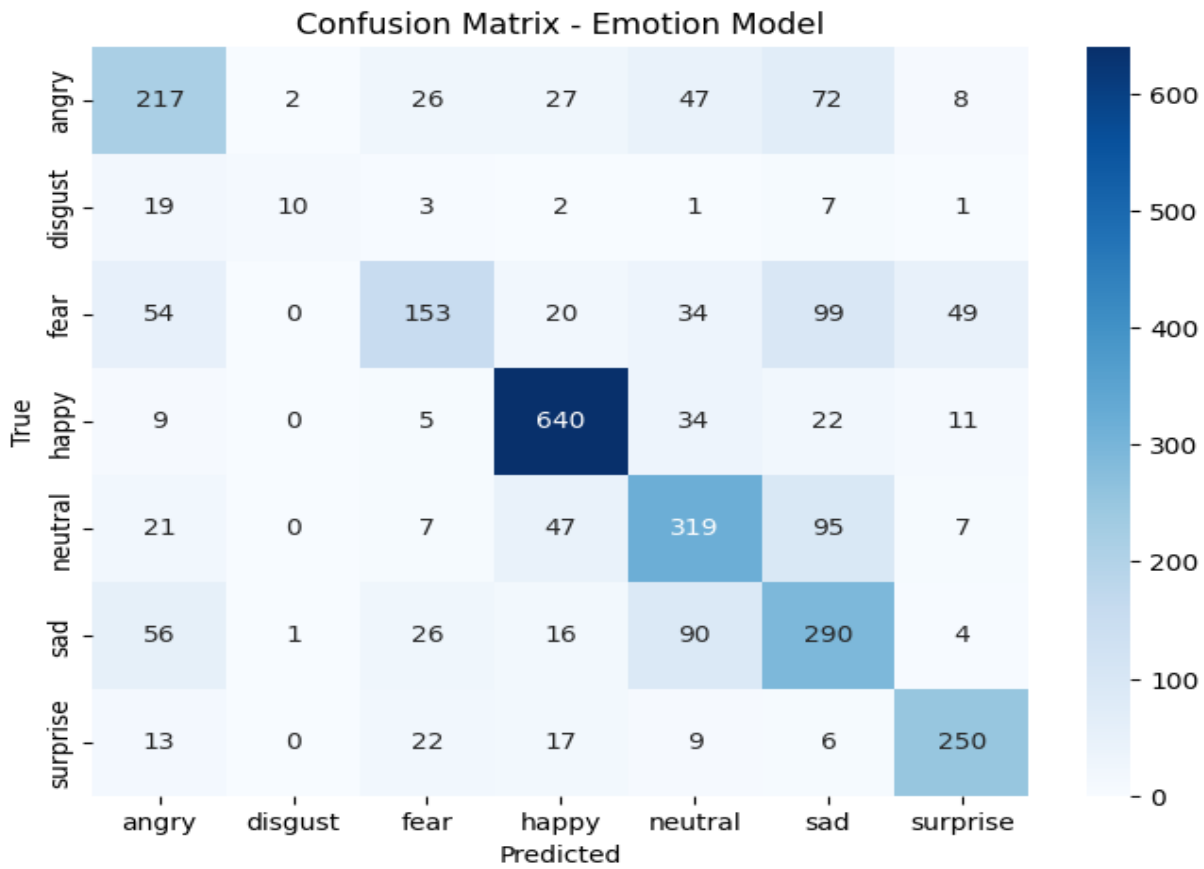


**Figure 4.1:** Confusion matrix - model predictions across different emotion classes.

## 4.2 ACCURACY AND TRAINING PERFORMANCE

The overall accuracy of the model reached approximately 85%, indicating strong reliability in its capacity to identify and categorize a broad set of emotional expressions. The training and validation accuracy curves displayed a stable upward trend, demonstrating that the model successfully learned meaningful patterns without significant fluctuation or instability. Similarly, the loss curves showed a steady decline and consistent alignment between training and validation losses, confirming effective generalization and minimal overfitting throughout the training process. These performance trends visualized in **Figure 4.2**, which presents the model's accuracy and loss progression over the training epochs confirm that the FER model is reliable and well-suited for real-time deployment within the recommendation system.

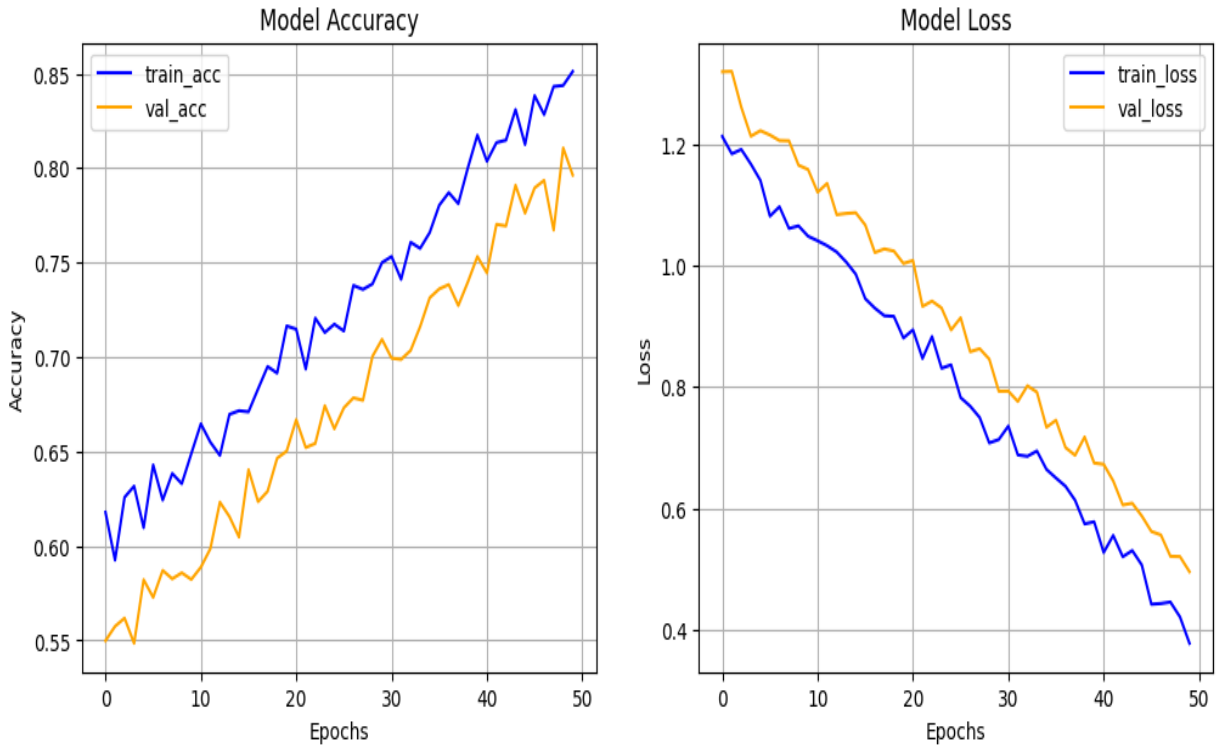


**Figure 4.2:** Model accuracy and loss curves over training epochs.

## 4.3 USER INTERFACE RESULTS

Muzly's interface is designed to offer a smooth and intuitive user experience, allowing effortless interaction across its main sections Home, Songs, About, Profile, and Authentication. The Home page, illustrated in Figure 4.3, serves as the primary entry point, enabling users to upload images and receive instant emotion-based music recommendations while showcasing the system's workflow and key features. The Songs page, shown in Figure 4.4, provides an accessible music library with search options for filtering by title, artist, language, or mood. The upload and analysis workflow is demonstrated in Figure 4.5 and Figure 4.6, where users can submit an image and view the detected emotion along with contextual insights. Once processed, the generated suggestions appear in the Recommended Songs section, depicted in Figure 4.7, presenting personalized music choices tailored to the user's emotional state. The Profile page, illustrated in Figure 4.8, allows users to manage personal information, access saved songs, and revisit previous recommendations, enhancing long-term personalization. the About page, shown in Figure 4.9, provides an overview of the system's purpose, underlying technologies, and design approach.

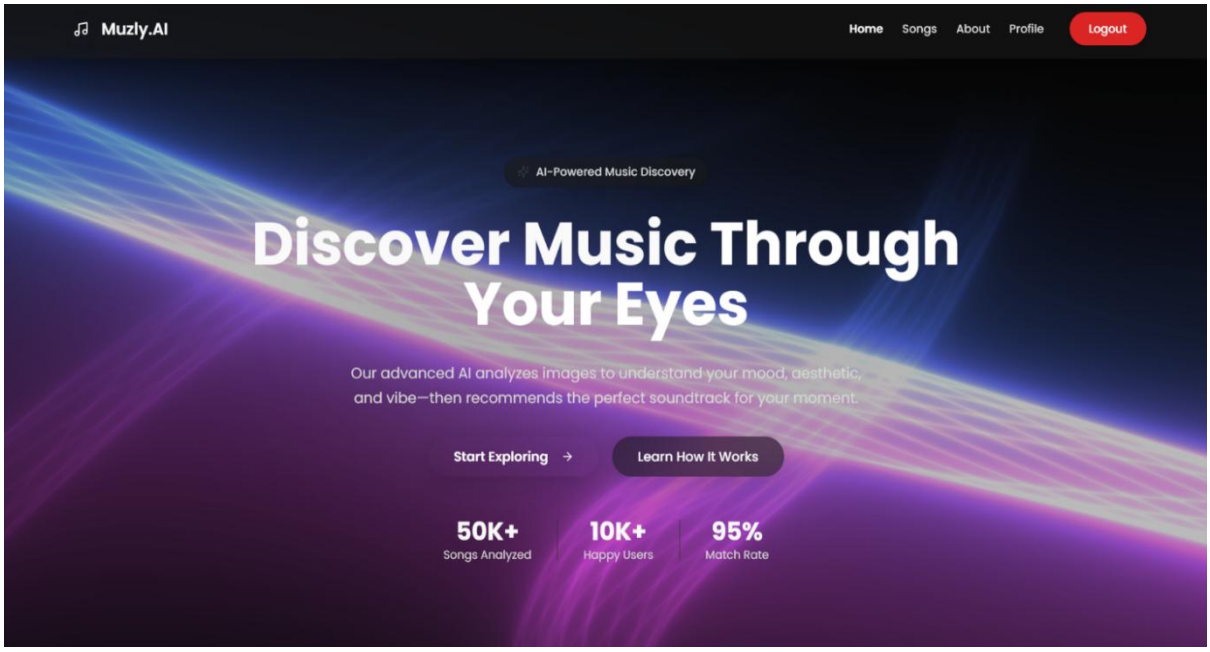


**Figure 4.3:** User Interface Image 1 (Home page)

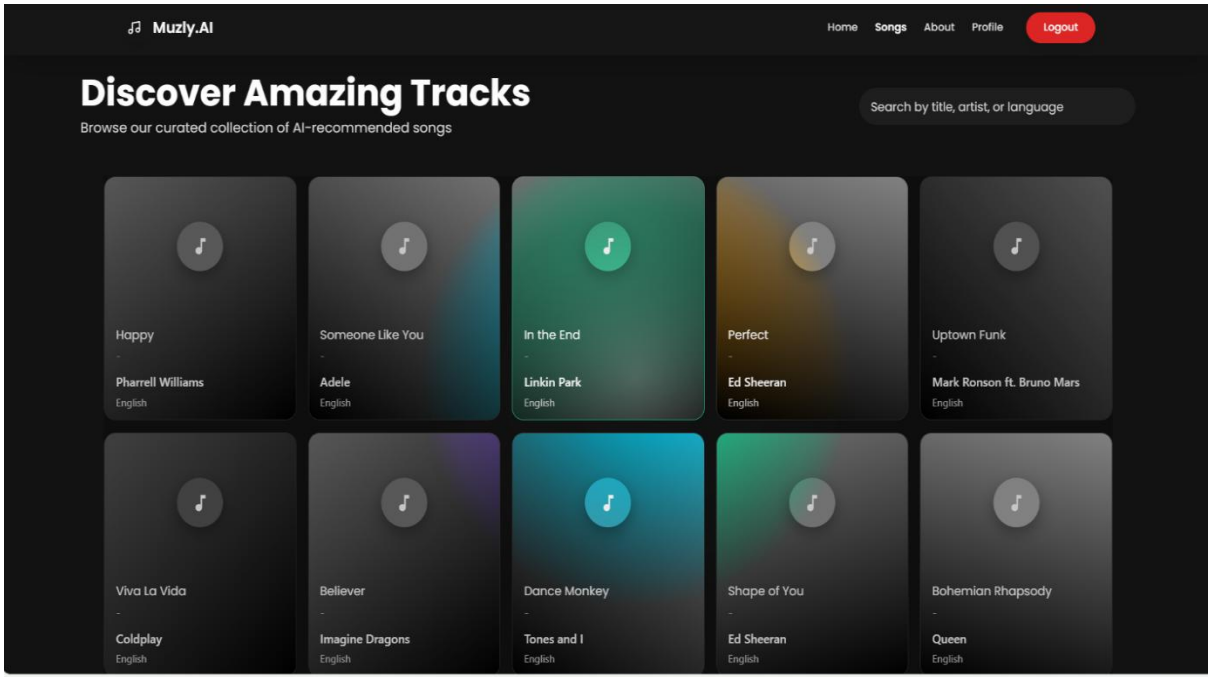


**Figure 4.4:** User Interface Image 2 (Songs page)

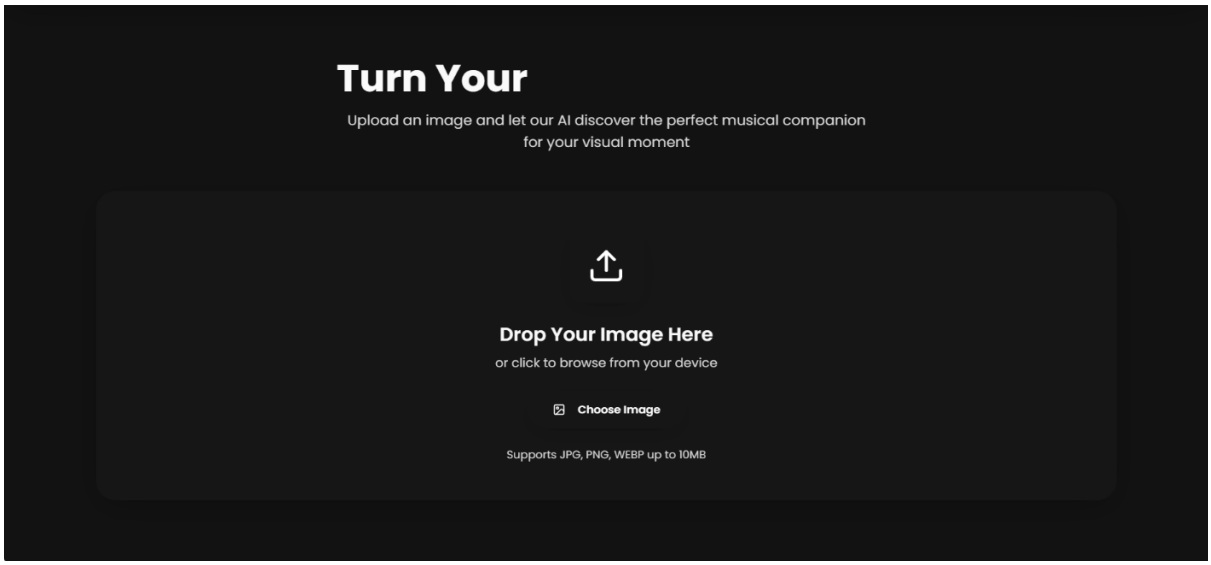


**Figure 4.5:** User Interface Image 3 (Upload section)

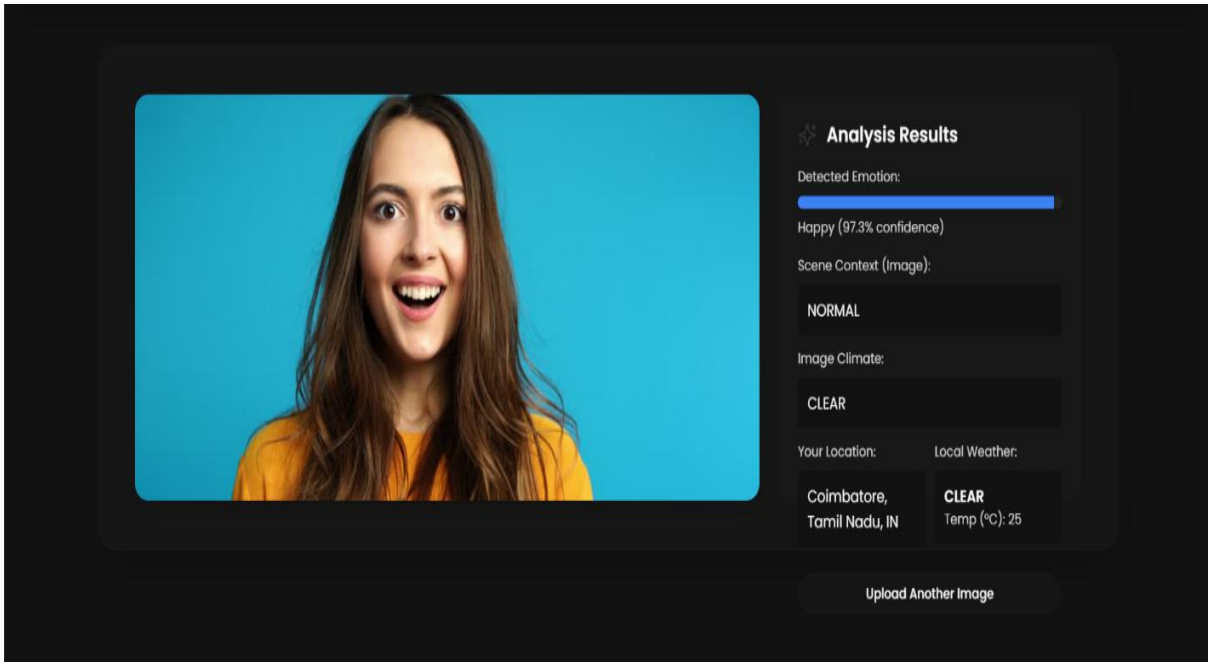


**Figure 4.6:** User Interface Image 4 (Analysis Section)

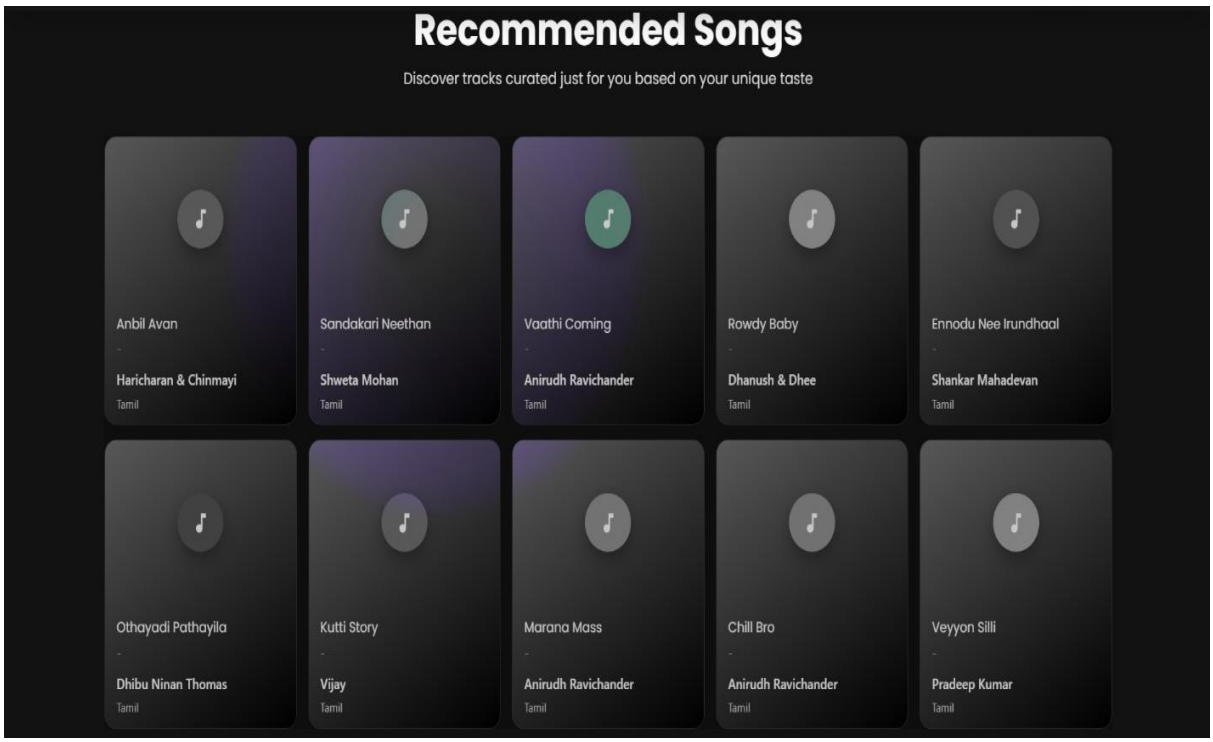


**Figure 4.7:** User Interface Image 5 (Recommended Songs section)

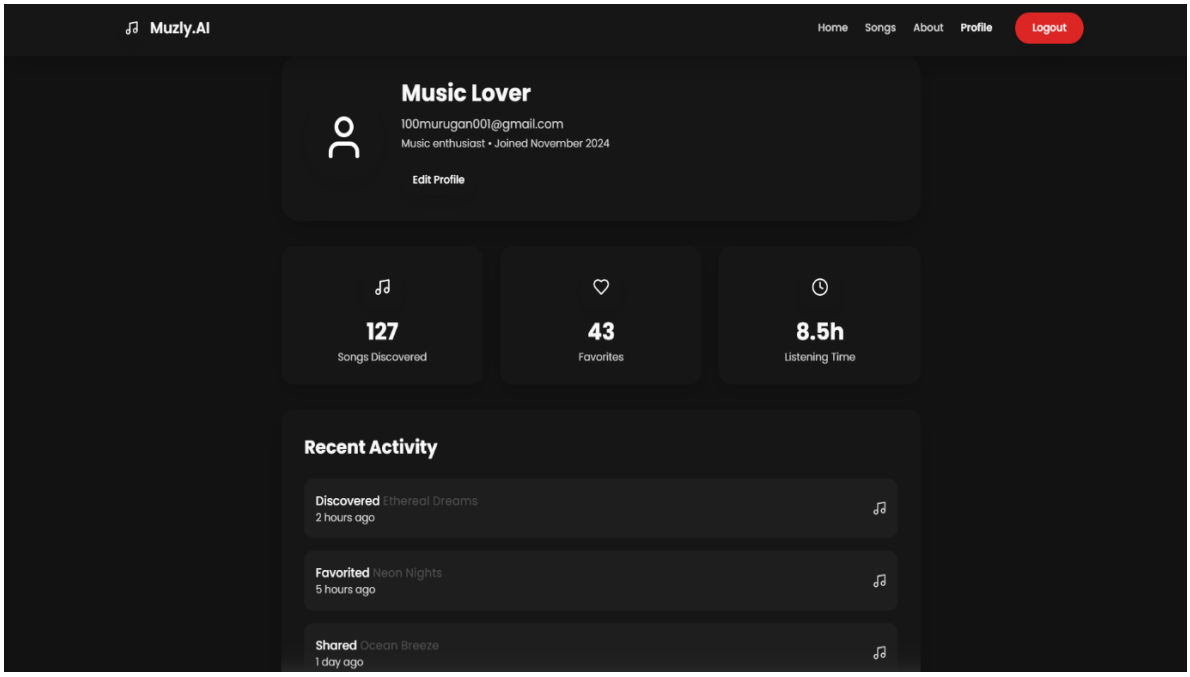


**Figure 4.8:** User Interface Image 6 (Profile page)

The results demonstrated that the developed FER-based music recommendation system performs reliably across its core components, from emotion detection to user interaction. The CNN model showed strong accuracy, stable training behavior, and dependable classification performance as confirmed through the confusion matrix and evaluation metrics, proving its suitability for real-time use. The system also maintained robustness under varied lighting and facial orientations, ensuring consistent results in practical scenarios. Furthermore, the integration of the FER module into the React interface enabled users to seamlessly upload images, receive instantaneous emotion analysis, and explore personalized song recommendations through an intuitive and visually organized UI. Overall, the results validate both the technical soundness and user-centered design of the system, confirming its effectiveness in delivering a smooth, engaging, and context-aware music recommendation experience.